



Inherent scatter in pool boiling critical heat flux on reference surfaces

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ABSTRACT

Evaluating boiling heat transfer enhancement depends on reliable reference values in the form of boiling curves and critical heat flux (CHF) values. Typically, the evaluation is performed in pool boiling conditions with water at atmospheric pressure. Literature includes a wide scatter in reference values, prompting this study to comprehensively evaluate boiling performance and CHF on reference surfaces to investigate the scatter's origin and set a definitive reference value. The study recorded 125 boiling curves and CHF values on nominally identical bare copper surfaces, establishing an average boiling curve and mean CHF value. Despite consistent experimental conditions, the recorded CHF values displayed significant variability with a mean CHF of $1112 \pm 102 \text{ kW m}^{-2}$ and a scatter from 902 kW m^{-2} (−19 % of average CHF) to 1339 kW m^{-2} (+25 % of average CHF). Using Rohsenow's correlation on the average boiling curve, a C_{sf} factor of 0.0151 was obtained. The acquired CHF data is proposed to serve as a foundational benchmark for future research in enhancing pool boiling heat transfer.

1. Introduction

Phase change heat transfer processes such as boiling have common industrial and household applications due to their high heat transfer intensity. Boiling heat transfer is already being used in industrial applications such as power plants [1–4], the chemical industry [5–7], water purification [7], in heat exchangers [8,9] and nuclear reactors [10,11]. Furthermore, boiling is also utilized for heat dissipation in high-performance electronic devices such as high-performance computers [12–15]. In many industrial applications, system safety and stability are critical, especially in applications such as nuclear reactors [16, 17]. The highest heat transfer intensity is mostly observed in the nucleate boiling regime, where individual bubbles form, coalesce and depart the boiling surface. With increasing heat flux, bubble crowding on the surface starts to lead to partial dryouts and eventually to the surface becoming completely covered by a vapor film, representing the transition into the film boiling regime, where heat transfer intensity can be several orders of magnitude lower.

In this regard, *critical heat flux* (CHF) is defined as the upper heat flux limit of the nucleate boiling regime. To maintain system safety and stability, critical heat flux should never be approached, let alone reached. If an applied heat flux reaches that limit (i.e., the CHF), a continuous vapor film covers the surface, leading to a drastic increase in thermal resistance and surface temperatures, likely causing the failure of

the cooled component or system with possible catastrophic consequences. In addition to CHF, two other parameters are of great importance in describing the heat transfer properties during nucleate pool boiling: the surface superheat at the onset of nucleation boiling (ONB) and the heat transfer coefficient (HTC). Surface superheat is defined as the difference between the surface temperature and the saturation temperature of the boiling fluid, while the HTC is defined as the ratio of heat flux and surface superheat. To show the relationship between heat flux and surface superheat, a so-called *boiling curve* is used and typically utilized to assess the heat transfer behavior under different conditions or on different boiling surfaces.

The critical heat flux, a crucial parameter in pool boiling systems, is influenced by a multitude of factors. Among these factors, surface characteristics play a significant role in determining the CHF value. Specifically, surface roughness [18,19], wettability [20,21], and geometry [22], along with the properties of the boiling fluid [19,23], contribute to the overall CHF behavior. Despite the importance of CHF, there is a significant amount of scatter in the reported values for reference surfaces, making it difficult to accurately predict and control this phenomenon [24]. A major contributing factor to this scatter is the complexity of the boiling process, which involves a variety of physical mechanisms, such as bubble nucleation [25,26], growth [27], departure [28], and coalescence [29], as well as the presence of surface roughness and impurities [30–32]. Another factor contributing to the scatter in

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reported CHF values is the lack of standardization in experimental procedures and conditions under which boiling tests are conducted. The current state of the experimental procedure of boiling tests in various studies typically involves degassing the surfaces prior to the experiment to eliminate entrapped gases within surface cavities. Additionally, researchers record the boiling curve up to the CHF incipience on selected surfaces and conduct multiple experimental runs to demonstrate the repeatability of the obtained results. All studies dealing with enhancement of boiling performance require a benchmark to compare the performance of functionalized surfaces. To this effect, a “bare” reference surface with comparable macroscopic geometry and dimensions to the functionalized surfaces, is utilized. Furthermore, the surface properties of the reference sample should resemble those of other samples before any specific treatments are applied to the latter. To ensure a fair comparison, the reference test must be conducted under identical conditions to the subsequent experiments involving enhanced surfaces. In the context of pool boiling, this entails maintaining consistency in parameters such as liquid bulk temperature, operating pressure, surface orientation, gravity, and the working fluid. The variability of the aforementioned parameters can lead to differences in the observed CHF values under similar experimental conditions, making it difficult to compare and generalize results across different studies.

A review of the literature reveals a notable discrepancy in the critical heat flux (CHF) values obtained on reference surfaces under comparable conditions. In line with the prevailing practice in the boiling research community, the reviewed studies predominantly employed copper surfaces as a preferred material. Copper's widespread popularity stems from its inherent advantages, such as high thermal conductivity, chemical stability, compatibility with diverse fluids, and ease of machining. Some critical heat fluxes for boiling of water on copper surfaces reported in the literature and observed under the same experimental conditions are 606 kW m^{-2} (Coursey & Kim [33]), 1568 kW m^{-2} (Bai et al. [34]), 870 kW m^{-2} (Cao et al. [35]), 1450 kW m^{-2} (Dai et al. [36]), 1110 kW m^{-2} (Hadžić et al. [23]), 1320 kW m^{-2} (Može et al. [6]), 775 kW m^{-2} (Alimoradi et al. [37]), 1178 kW m^{-2} (McNamara et al. [38]), 1050 kW m^{-2} (Gupta et al. [39]), 985 kW m^{-2} (Ranjen et al. [40]), 956 kW m^{-2} (Hsu et al. [41]), 1100 kW m^{-2} (Girad et al. [42]), 898 kW m^{-2} (Mehdikhani et al. [43]). The aforementioned list represents just a small sample of the published CHF data, but clearly illustrates that the scatter in reported CHF values is huge, ranging from 600 kW m^{-2} to 1600 kW m^{-2} for saturated boiling of water on bare copper surfaces at atmospheric pressure [44–47]. Furthermore, we analyzed the effect of atmospheric pressure change on the CHF, wherein no distinct trend in CHF was observed. Moreover, the observed inconsistency and scatter of the CHF values on reference surfaces not only introduce complexity in the comparison of experimental findings regarding heat transfer performance and CHF enhancement with other researchers owing to significant disparities in values recorded on reference surface, but also represents a potential safety hazard in the event that boiling surfaces display a lower critical heat flux (CHF) values than expected (i.e., found in the literature) under practical operating conditions. From this, the core question of this study arises: over what range will the CHF values, recorded under identical conditions, be inherently scattered due to complexities and randomness of the boiling process?

The present investigation addresses the aforementioned concerns by examining the impact of distinct parameters, namely wettability, reuse of surfaces, and the number of experimental runs, on the critical heat flux (CHF) during pool boiling. A total of 25 bare copper surfaces were prepared and tested in nominally identical way, i.e., by sanding and subsequent cleaning, with five boiling curves recorded on each surface up to the CHF incipience, leading to the acquisition of a total of 125 boiling curves and 125 CHF values. The pool boiling tests were carried out with double-distilled water under saturated conditions at atmospheric pressure, representing the most common conditions used in literature. All 125 boiling curves were averaged into a single average boiling curve. The obtained data was also evaluated using the

Rohsenow's correlation. Furthermore, the influence of surface wettability prior to the boiling test, the number of experimental runs, and the frequency of (re)use of a single sample for experiments on CHF values were analyzed. Normalized data was subjected to linear regression to identify the effect of each parameter on CHF, and analysis of variance (ANOVA) was employed to determine whether the effect of each parameter was statistically significant. Lastly, a comparison of experimental data and heat transfer enhancement reported by other researchers is presented using our findings regarding the average boiling curve and CHF value to highlight the possible discrepancies in reporting heat transfer performance in a relative manner.

2. Material and methods

2.1. Sample

The manufacturing process for bare copper samples (electrolytic purity, $>99.9\% \text{ Cu}$) with 14 mm diameter involved a three-step procedure: (1) sanding the surface, (2) cleaning the surface, and (3) gluing and mounting the sample. The sanding process was carried out using P600, P800, P1200, and P2000 abrasive papers in sequence. Each sample was sanded on each sandpaper in variable directions for approximately 30 s. A surface roughness of $0.23 \pm 0.02 \mu\text{m}$ was achieved after sanding, which was measured using a Mitutoyo SJ-310 surface roughness tester. Following this, all samples were cleaned with 2-propanol and in a UV ozone cleaner (Ossila Ltd, Sheffield, UK) to remove any contaminants from the surface.

The sample being tested was then mounted onto a copper block and fixed with an insulative PEEK holder. To prevent parasitic boiling and seal the gap at the edge of each sample, a small amount of flexible epoxy resin (Duralco 8453-Cotronics) was applied, which was hardened in an oven at 90°C for 90 min.

Temperature of the sample was measured with three K-type thermocouples, providing information about the axial temperature gradient, and allowing the calculation of heat flux. More detailed information about the samples can be found in Fig. S2 of the Supporting Information, Section 1.

2.2. Pool boiling performance evaluation

Pool boiling at atmospheric pressure was conducted on the experimental setup illustrated in Fig. S1, Supporting Information, Section 1. The glass boiling chamber with an inner diameter of 60 mm was filled with 200 mL of water. Saturation temperature was achieved and maintained with an immersion heater, while the sample was heated with three cartridge heaters inside the heating block. The water temperature was measured with two submerged K-type thermocouples at different heights inside the chamber. A condenser, cooled by a secondary water loop, was mounted on top of the assembly to keep the water level steady during experiments, while the top of the condenser was open to the environment which prevented overpressure from building up in the boiling system. Specifically, we also selected and installed a condenser and fitting with a large internal diameter (17.5 mm or more), that prevents flooding conditions and allows continuous vapor/liquid countercurrent throughout the entire nucleate boiling regime including the CHF. Measurements were therefore performed at ambient atmospheric pressure. All temperature signals were recorded as raw voltages at 10 Hz, using a DAQ device (KRYPTONi-8xTH) and Dewesoft X3 software. Additional information about the experimental setup is available in our previous work [23,48,49].

2.3. Measurement procedure

In this study, a total of 125 experimental runs were conducted to measure the pool boiling performance of 25 bare copper samples. The experiments were divided into three groups. The first group consisted of

12 samples, which underwent boiling tests (five runs on each sample). Subsequent to these tests, six samples underwent thorough surface analysis (SEM and EDS analyses), while the remaining six underwent boiling tests in the second group. Additionally, one of the samples that underwent surface analysis was also included in the boiling tests. Consequently, in the second group, seven samples were designated for boiling tests. Following the completion of boiling tests for the seven samples in the second group, one sample experienced thermocouple failure. As a result, out of the original seven samples in the second group, only six were available for further testing in the subsequent, third group. One of the samples from the second group was not tested further due to thermocouple failure. A schematic view of the experimental procedure is provided in Fig. 1.

The measurement protocol for each tested sample in this study involved degassing water in the boiling chamber to obtain pure boiling liquid free of non-condensable gases. This was achieved by switching on the immersion heater for 30 min followed by switching on the cartridge heaters for 15 min to maintain boiling at approximately 250 kW m^{-2} . This procedure yielded a degassed surface without entrapped vapor or gas, to avoid early onset of nucleate boiling. The degassing process also served for thermal stabilization of the sample, which was exposed to temperatures of approximately $100\text{--}150^\circ\text{C}$ for 45 min. After degassing, the temperature was reduced to under 90°C to ensure the condensation of any leftover vapor pockets. Following this, the measurement was performed at atmospheric pressure and a stable saturation temperature. When recording a boiling curve, the heat flux was increased at a rate of less than $0.2 \text{ kW m}^{-2} \text{ s}^{-1}$ during the natural convection regime and at a rate of less than $2 \text{ kW m}^{-2} \text{ s}^{-1}$ during nucleate boiling. This dynamic measurement approach was evaluated and validated in a previous study [50]. Each boiling curve was recorded until CHF, after which the cartridge heaters were turned off and the sample was left to cool down before initiating the next run. The CHF onset was detected through a momentary steep rise in sample and surface temperature as observed from the thermocouple readings. Each recording took approximately 45 min, while 5 consecutive pool boiling curves on each sample required 5 h to complete.

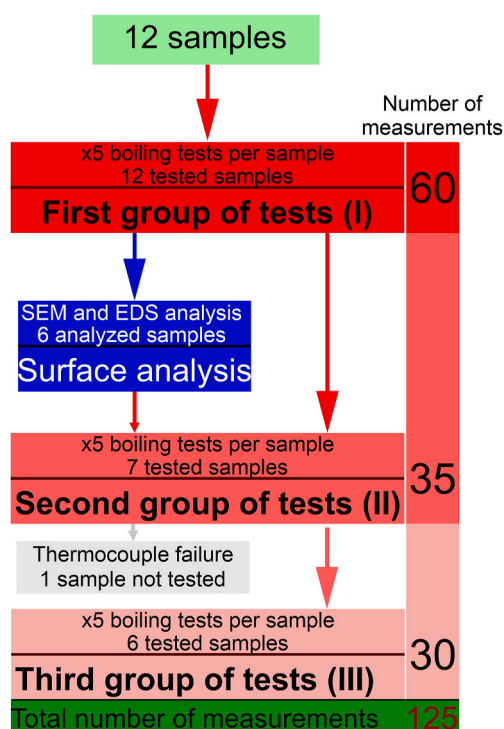


Fig. 1. Schematic view of the experimental procedure.

2.4. Data reduction and measurement uncertainty

Altogether, five temperatures were measured during the experiment – three inside the sample and two in the water. From these the heat flux, surface superheat and heat transfer coefficient were calculated and recorded in real time with the DewesoftX software, while the post-processing was conducted with Mathworks MATLAB R2021a. More details are available in our previous work [23,49,51].

Briefly, temperature-dependent thermal conductivity was taken at the average sample temperature, obtained as the arithmetic mean of three thermocouple measurements. Heat flux was calculated with Fourier law of one-dimensional heat conduction, using the temperature gradient obtained from the thermocouple measurements. Surface superheat was obtained as the difference between sample surface and water temperature. The latter was taken as the arithmetic mean of both submerged thermocouple measurements, while the surface temperature was estimated with the measurement closest to the surface (upper sample thermocouple) and the heat flux. Heat transfer coefficient was obtained by dividing the heat flux with the surface superheat.

Measurement uncertainty was evaluated following recommendations of a previous study [24]. The standard measurement uncertainty values of the three main parameters (i.e., heat flux, surface superheat, and heat transfer coefficient) are dependent on operating conditions. Therefore, uncertainty analysis was conducted at two heat flux levels, at 200 kW m^{-2} and at a higher value of 1000 kW m^{-2} . At 200 kW m^{-2} , the relative standard measurement uncertainty of heat flux is 10.91–11.03 %, the standard measurement uncertainty of surface superheat is 0.78 K and the relative standard measurement uncertainty of the heat transfer coefficient is 11.44–13.44 %. At 1000 kW m^{-2} , the aforementioned standard measurement uncertainties are 4.48–5.33 % for heat flux, 1.39 K for surface superheat and 7.86–10.18 % for the heat transfer coefficient. The detailed analysis of measurement uncertainty is shown in Table S1, Supporting Information, Section 1.

2.5. Surface analysis

Sample surface was analyzed with scanning electron microscopy (SEM; Quattro S, ThermoFisher Scientific Inc., Waltham, MA, USA) at an accelerating voltage of 10 kV in backscattered electron (BSE) and secondary electron (SE) mode. SEM images of bare copper surfaces were obtained before (left part of Fig. 2) and after (right part of Fig. 2) boiling tests at various magnifications, as shown in Fig. 2. The bare copper surfaces before boiling exhibited only tool marks from abrasive sand-paper treatment, and the same case was observed for the bare copper surfaces after boiling tests. EDS analysis of the surface was also conducted, in addition to SEM analysis. The findings revealed that the oxygen content was higher on surfaces after boiling tests than before. This observation is consistent with earlier studies, which have shown that copper oxides are formed and transformed on surfaces due to CHF incipience. Detailed EDS analysis of the surfaces before and after boiling is shown in Fig. S4 and can be found in the Supporting Information, Section 2.

Before and after boiling, the wettability of each sample was gauged with sessile droplet measurements of apparent contact angle on a goniometer device and its proprietary software (Ossila Ltd., Sheffield, UK). 20 μL were deposited on different locations on each sample. Before boiling, the mean apparent contact angle for 25 tested samples was $73.8^\circ \pm 14.5^\circ$, while the average apparent contact angle immediately after boiling experiments for all samples was $62.4^\circ \pm 9.5^\circ$. Fig. S6, Supporting Information, Section 3, displays the contact angle measurements for each tested sample.

After the boiling process, the surface roughness of the tested samples was measured using Mitutoyo SJ-310. The average surface roughness of the first group of boiling test surfaces was $0.20 \mu\text{m}$, with a standard deviation of $0.02 \mu\text{m}$. In addition, five more samples were prepared with the same methodology as the boiling samples, and their surface

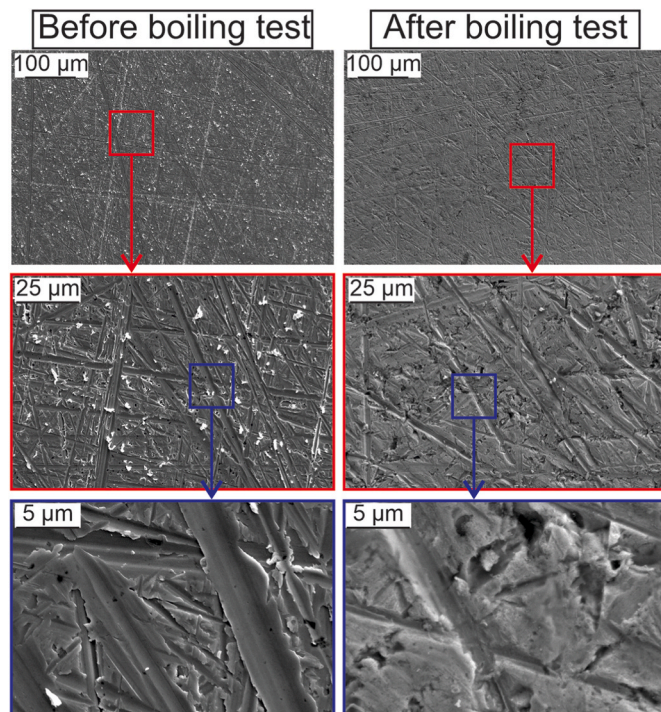


Fig. 2. SEM images of reference copper surface before and after boiling tests.

roughness was also measured to obtain the surface roughness value before the boiling tests. The mean surface roughness of these additional samples was $0.23 \mu\text{m}$, with a standard deviation of $0.02 \mu\text{m}$. Detailed roughness analysis of the surfaces before and after boiling is shown in Fig. S5 and can be found in the Supporting Information, Section 3.

3. Results

3.1. Evaluation of pool boiling heat transfer performance

The study evaluated the boiling performance on 25 bare copper surfaces with five experiments conducted on each surface, resulting in 125 boiling curve measurements up to the critical heat flux (CHF) for all surfaces. All measurements were performed at atmospheric pressure under saturated conditions using double-distilled water. Boiling curves were generated to display the relationship between the heat flux and the surface superheat. Heat flux and superheat values were extracted at heat flux increments of 50 kW m^{-2} and averaged at each heat flux to obtain the average point (i.e., average surface superheat for a given heat flux), culminating in the average boiling curve shown in Fig. 3 together with the standard deviation of individual averaged datapoints and the scatter of the CHF values.

The calculated average CHF value from the 125 CHF values recorded on a total of 25 copper surfaces was 1112 kW m^{-2} with a standard deviation of 102 kW m^{-2} . The CHF data exhibited a scatter as shown in upper part of Fig. 3, with the lowest and highest CHF values being 902 kW m^{-2} and 1389 kW m^{-2} , respectively. The lower part of Fig. 3 shows a more precise view of the scatter of the critical heat flux. The difference of 486 kW m^{-2} between the lowest and highest CHF value (representing 44 % of the mean CHF value), along with the standard deviation of 102 kW m^{-2} , is well outside the measurement uncertainty of the experimental apparatus. The highest measurement uncertainty recorded during one experimental run is 52.1 kW m^{-2} at a heat flux level (1100 kW m^{-2}) near average critical heat flux calculated from the 125 measurements. The standard deviation of the CHF is therefore double compared to the highest recorded measurement uncertainty at the same heat flux level. This suggests that the scatter of CHF values was due to the

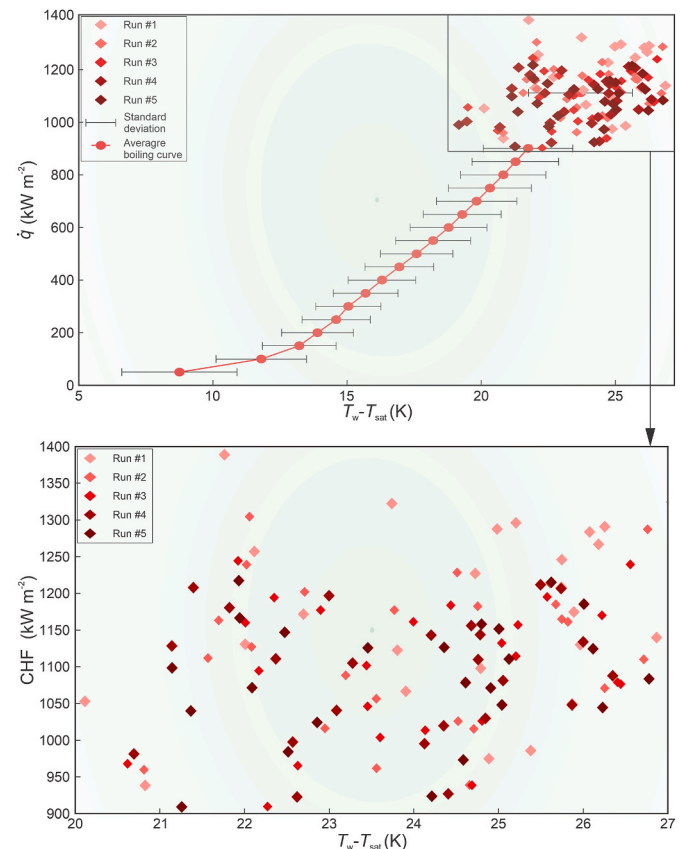


Fig. 3. Average boiling curve calculated from 125 measurements (upper part) and scatter of the CHF datapoints (lower part).

stochastic behavior of the boiling process on nominally identical surfaces, prepared and tested using the same preparation procedure, and surely not due to measurement uncertainty. Additionally, the average boiling curve coupled with a maximum and minimum values of superheat temperature in the heat flux range between 50 kW m^{-2} and the CHF was shown in Fig. S7 in Supporting Information, Section 4.

Fig. 4 shows the heat transfer coefficients at specific heat flux levels ($200, 400, 600, 800 \text{ kW m}^{-2}$) for 125 datasets. The heat transfer coefficients are ordered by experimental runs from the first to the fifth run on each surface, which were performed sequentially. A shift in the boiling curve towards higher or lower superheat temperature on subsequent experimental runs results in an increase or decrease in HTC on the specific bare copper samples. This is consistent with previous studies that have also shown that the HTC can decrease or increase with each subsequent experimental run [5,52]. The overall statistical analysis showed that the mean HTC value (calculated for one experimental run using measurements on 25 surfaces) decreases with each subsequent run. The deterioration of the mean HTC value from the first to the fifth run increases with increasing heat flux. At a heat flux of 200 kW m^{-2} , the deterioration of the mean HTC value from the first to the fifth run is approximately 11 %, while the deterioration of the mean HTC value from the first to the fifth run at a high heat flux is lower (at 800 kW m^{-2}) and is approximately 3.8 %. Additionally, the outliers from sets of data shown in Fig. 4 are determined as data that are outside of 1.5 times the interquartile range.

In most studies dealing with long-term testing of surfaces, the effect of surface aging on boiling heat transfer parameters is considered a significant factor when the boiling process was conducted on that surface for more than 20 h [53–55]. In this study, each experimental run lasted about 1 h, resulting in a 5-h boiling test. Therefore, the observed slight deterioration of the heat transfer coefficient cannot be attributed

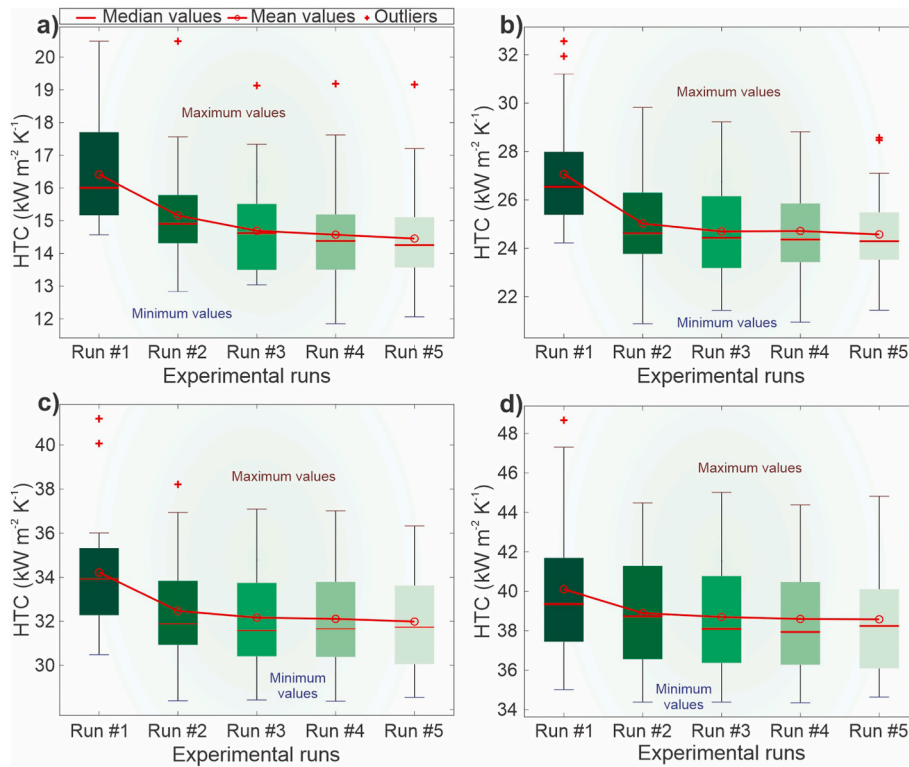


Fig. 4. The change of heat transfer coefficients with increasing the number of experimental runs at heat flux levels: a) 200 kW m^{-2} , b) 400 kW m^{-2} , c) 600 kW m^{-2} and d) 800 kW m^{-2} .

to the aging effect of the surfaces.

4. Discussion

4.1. Evaluation of experimental data using Rohsenow's correlation

The Rohsenow's correlation has gained significant attention in the scientific community owing to its robust capability in predicting the heat transfer coefficient in boiling heat transfer applications with reasonable accuracy. Moreover, it has been predominantly employed for the purpose of validating experimental setups and the results obtained from them [56–60]. The correlation's validation through extensive experimental data and favorable comparisons with other established correlations serve to reinforce its credibility and wide applicability, rendering it as one of the most extensively utilized correlations in the field. Therefore, the obtained average boiling curve from 125 experimental measurements on 25 bare copper surfaces was compared to the boiling curve obtained by values from the Rohsenow's correlation [61] given by:

$$\frac{c_{pL}\Delta T_{sat}}{h_{LG}} = C_{sf} \left\{ \frac{\dot{q}}{h_{LG}\mu_L} \left[\frac{\sigma}{g(\rho_L - \rho_G)} \right]^{0.5} \right\}^m \left(\frac{c_{pL}\mu_L}{k_L} \right)^n \quad (1)$$

$$\dot{q} = h_{LG}\mu_L \left[\frac{g(\rho_L - \rho_G)}{\sigma} \right]^{0.5} \left\{ \Delta T_{sat} \frac{c_{pL}}{h_{LG}C_{sf}} \left(\frac{k_L}{c_{pL}\mu_L} \right)^n \right\}^{1/m} \quad (2)$$

Thermophysical properties of water boiling at saturation temperature and atmospheric pressure as used in this study are shown in Table S2, Supporting Information, Section 5. The exponents m and n were set to 0.33 and 1, respectively, for the combination of water and copper surfaces, following previous studies [57,62]. The surface factor constant (C_{sf}) in the Rohsenow's correlation was determined by fitting the correlation data to the average experimental data of all 125 measurements in the heat flux range between 50 kW m^{-2} and the average CHF. The obtained value for C_{sf} was 0.015053, with an R-squared value of 0.9946, indicating a strong fit. The experimental boiling curve was

found to agree well with the boiling curve predicted by the Rohsenow's correlation, and the determined C_{sf} value falls within the range of values reported in previous studies for the same liquid-surface combination [57,58,62]. Comparison of the calculated average boiling curve to the boiling curve obtained by Rohsenow's correlations is shown in Fig. 5a.

Furthermore, experimental boiling curve furthest to the left and right of the average boiling curve, representing the lowest and highest recorded superheat temperatures at each heat flux level from 50 kW m^{-2} to CHF, were also fitted to the data obtained from the Rohsenow's correlation. The resulting C_{sf} values were 0.01204 and 0.01708, respectively, with R-squared values of 0.9513 and 0.9417. The fit of the boiling curves obtained using the Rohsenow's correlation to the leftmost and rightmost experimental boiling curves can be found in Fig. S8 of the Supporting Information, Section 5. Fig. S9 and Fig. S10, Supporting Information, Section 5, show the scattering of C_{sf} values, calculated for each recorded boiling curve, over CHF and superheat temperatures recorded at CHF, respectively.

Moreover, in Fig. 5b, we present a comparative analysis of predicted surface superheats in relation to the corresponding measured surface superheat data, with a confidence level of 95 %. Notably, the predicted data points at higher surface superheat lay within the 95 % confidence interval, indicative of reliable predictive performance in this regime. However, deviations from the confidence level become evident at lower surface superheats, manifesting as outliers in the data.

These observed outliers at lower surface superheats can be attributed to the inherently scattered and stochastic nature of onset of nucleate boiling. The complex interplay of microscale phenomena during nucleation events renders the predictions more challenging in this regime, leading to the observed deviations from the confidence interval.

4.2. Regression analysis of the collected CHF data

Two main contributions to variations in the heat transfer intensity and the CHF value (when keeping all other parameters such as the size,

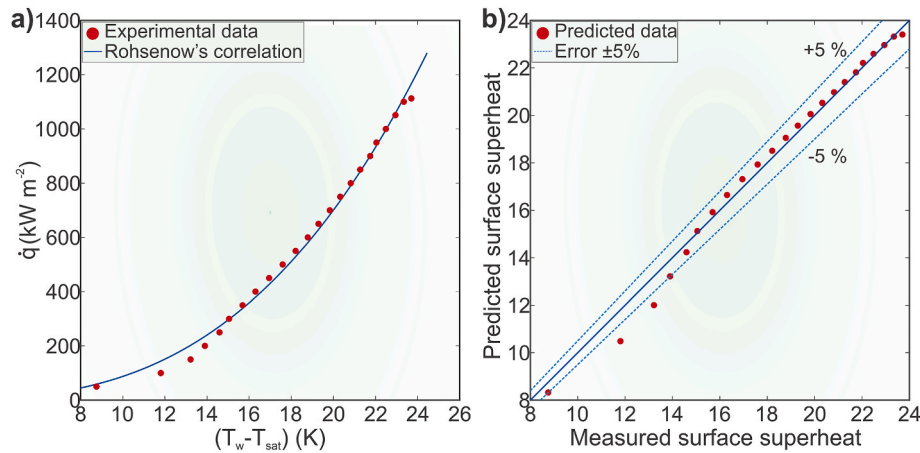


Fig. 5. a) Comparison of the calculated average boiling curve to the boiling curve obtained by Rohsenow's correlations and b) comparison of predicted surface superheat data over measured surface superheat data with confidence level of 95 %.

orientation and shape of the sample and testing conditions the same) are surface topography and surface wettability. Effects of surface topography include the influence of roughness, the size of the rough peaks or cavities, or the distance between them on the CHF values [31,63,64], which is attributed to the higher population density of cavities (which can serve as potential nucleation sites) on the surface. Previously, Može et al. [24] studied the statistical influence of roughness on CHF by using different data from the literature. They showed that roughness has a statistically significant influence on CHF. CHF increased with increasing surface roughness, which was also confirmed by a positive regression coefficient. However, the study also notes that many papers fail to report the roughness of the samples or report it through different metrics, making comparisons practically impossible.

In this study, surface topography parameters were measured, specifically the surface roughness. Five samples were directly prepared for roughness measurements, and the results showed an arithmetic average roughness (R_a) of $\sim 0.23 \mu\text{m}$ on all tested samples. Additionally, roughness was tested on 12 samples after the first series of boiling tests, with the measurement showing an arithmetic average roughness of $\sim 0.20 \mu\text{m}$ on all tested surfaces. SEM analysis of the surfaces before boiling also confirmed that there were no significant differences in surface topography and morphology. Detailed SEM analysis can be found in the Supporting Information, Section 2. The measurement and comparison of surface topography parameters showed that the surfaces tested within this study are indeed nominally identical roughness-wise (Fig. S5 in the Supporting Information; Section 3), with negligible differences stemming from the sample preparation methodology, indicating that other factors must impact the scatter in CHF values.

The second parameter that influences the variation of CHF values, as previously mentioned, is surface wettability. Prior to the boiling tests, measurements of surface wettability exhibited noteworthy discrepancies, with a maximum recorded value of 95.4° and a minimum recorded value of 45° . The average surface wettability value among the 25 surfaces was 73.8° (Fig. S6 in the Supporting Information, Section 3 shows recorded contact angle values on each tested surface). Drawing from observed variations in wettability and from earlier research indicating the impact of surface wettability on CHF [65,66], we conducted a statistical analysis to assess the significant influence of wettability on CHF.

Regression analysis of the collected data was based on (i) the wettability of the surface prior to boiling experiments (i.e., its static contact angle, range of values: 45° – 95°), (ii) the number of experimental runs conducted on the surface since its fabrication (range of values: 1–5) and (iii) the number of times the surface was used for an experimental set of 5 measurements (range of values: 1–3). With the latter parameter, the aim was to exclude possible influences of thermocouple calibration,

different material properties or minute deviations in sample dimensions. For each set of 5 measurements, the sample was freshly pretreated, effectively using it as a new surface. The experimental measurements were classified into experimental groups by the number times of using samples is used for boiling tests, meaning there are: (Group I) samples not previously used for boiling experiments, (Group II) samples previously used for one series of 5 tests, and (Group III) samples previously used for two series of 5 tests each (see Fig. 1). A linear regression model was applied to the normalized data to examine the effect of each parameter on CHF on the bare copper surface. Analysis of variance (ANOVA) was utilized to determine if the influence of each parameter was statistically significant.

A linear regression model was constructed using a dataset of 125 experimental data points, which included the CHF value as the output dependent variable while contact angle before boiling, experimental group number, and run number were independent variables. All parameters were normalized between zero and one. The resulting model had 121 degrees of freedom and an F-value of 7.59 ($p = 0.001$) compared to a constant model.

The analysis of variance (ANOVA) indicated that the number of experimental runs on each sample was statistically significant factor affecting CHF (p -value < 0.05). CHF had a negative trend, i.e., it deteriorated with increasing number of experimental runs conducted on a sample.

As mentioned before, the wettability of the surface before boiling, evaluated through the contact angle measurements, was also expected to influence CHF, based on findings of previous studies [51,67,68]. However, the ANOVA test performed in this study did not find a statistically significant effect of wettability on copper surfaces before boiling tests on CHF (p -value = 0.667, above the significance threshold of 0.05). Prior research has demonstrated that surface wettability significantly affects critical heat flux (CHF). The range of contact angle in previous studies focusing on the effect of the wettability of the surface of boiling performance and critical heat flux was from 0° to 165° [69,70]. Specifically, hydrophilic surfaces with lower contact angles generally lead to an increase in CHF and a shift in the boiling curve towards higher superheats, while hydrophobic surfaces with higher contact angles tend to shift the boiling curve towards lower superheats and increase the heat transfer coefficient (HTC) [33,71,72]. However, in the present study, the range of contact angles recorded was much narrower, ranging from 45° to 95° . This reduced variability in the contact angles may explain why the statistical analysis failed to reveal a significant impact of wettability on CHF. Additionally, the range of contact angles measured after the boiling tests were even narrower (45° – 85°), indicating that bare copper surfaces may not exhibit substantial differences in wettability between them, thereby minimizing the effect on CHF in this particular case.

Detailed results of the regression model coefficients, the equation of the model, along with the ANOVA analysis are presented in the Supporting Information, Section 6. Furthermore, the CHF data was found to fit the normal distribution, as shown in Fig. S11 in the Supporting Information, Section 6.

Fig. 6 displays the CHF values plotted against the experimental run number (i.e., the sequential number of the current test since the surface was freshly prepared). Each experimental run comprises of 25 CHF values. The highest CHF mean value is observed in the first group, followed by a decrease in the mean value until the lowest value is reached in the fifth experimental run, as evidenced by the negative regression coefficient with value of -107.59 . This decline in the CHF value can be attributed to the change in the chemical structure of the boiling surface caused by the onset of CHF and the substantial increase in surface temperature during film boiling following CHF. Previous studies have demonstrated that during the initiation of CHF on copper surfaces and the transition to film boiling (during which the surface temperatures increase to 250°C or more), incomplete annealing of the surface occurs, transforming stable copper oxides into less stable ones [52]. This modification in surface chemistry and morphology leads to a decrease in CHF in subsequent experimental runs [6,52]. Our findings are consistent with this observation, as we observed a decline in CHF from the first to the fifth experimental run by an average of 7.6 %. EDS analysis (refer to Fig. S4 in the Supporting Information, Section 2) revealed an increase in oxide species on the surface, as evidenced by a slight percentage increase in oxygen. Therefore, our results suggest that changes in the surface chemistry due to CHF incipience followed by film boiling have a negative impact on CHF in subsequent experimental runs.

Furthermore, the impact of the number of times each sample was used for boiling experiments on critical heat flux (CHF) was also investigated. As previously mentioned, three experimental groups were used. It was observed that the mean value of CHF first slightly increases and then decreases with the increasing number of times an individual sample was used for boiling experiments. The ANOVA test performed in this study did not find a statistically significant effect on the number of times each sample was used for boiling experiments on critical heat flux with a p -value of 0.063, which is above the significance threshold of 0.05. The study tested 12 samples in Group I, 7 samples in Group II, and 6 samples in Group III, while the same 6 samples tested in all three groups were used to analyze the impact of the number of times each sample was used for boiling experiments on critical heat flux. The change in critical heat flux over the experimental group is shown in Fig. 7. However, some samples exhibited increased CHF with an increasing number of boiling tests, while others exhibited decreasing or

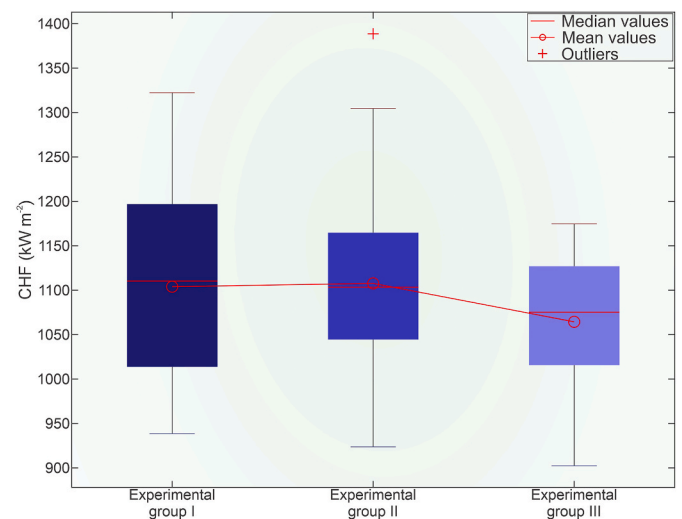


Fig. 7. The change of the critical heat flux over an experimental group.

random changes in CHF. More detailed information on the results of the ANOVA and regression analysis can be found in the Supporting Information, Section 6. The observation of the influence of the number of times copper surfaces were used for boiling tests on CHF per separate sample is depicted in Fig. S12 of the Supporting Information, Section 6.

Based on the findings presented in Fig. 7, an increase of approximately 0.3 % in CHF can be observed from the first to the second experimental group, and then a decline in CHF of approximately 4 % from the second to the third experimental group. Additionally, the outliers from sets of data shown in Fig. 7 are determined as data that are outside of 1.5 times the interquartile range. The observed trend of CHF over the experimental group is inside of measurement uncertainty, i.e., below 6 %. This agrees with findings of ANOVA analysis that showed that the impact of the number of times each sample was used for boiling experiments on CHF is not statically significant (p -value = 0.063, above the significance threshold of 0.05). Therefore, a potential factor that could contribute to the slight variations in CHF among the tested samples is the subtle alterations in surface chemistry and morphology that may occur during the 5-h boiling duration. These slight differences have the potential to influence the observed variations in CHF. Overall, it is important to acknowledge that boiling is an inherently stochastic process, influenced by numerous factors that were not specifically examined in our study. Consequently, there exist additional variables that could lead to slightly different CHF values when reusing samples for boiling tests.

4.3. The evaluation of the CHF scatter during the first experimental run

The calculated average CHF value from the 25 CHF values recorded on a total of 25 copper surfaces obtained during the first experimental run was 1163 kW m^{-2} with a standard deviation of 102 kW m^{-2} . The CHF data exhibited a scatter during the first experimental run as shown in Fig. 8, with the lowest and highest CHF values being 939 kW m^{-2} and 1389 kW m^{-2} , respectively. The difference of 450 kW m^{-2} between the lowest and highest CHF value (representing 39 % of the mean CHF value), along with the standard deviation of 117 kW m^{-2} . Further, it was also observed the difference in superheat temperature at which critical heat flux was occurring. The highest and lowest superheat temperatures obtained were 27.7 K and 20.1 K , respectively, alongside with average value of 24.5 K . The difference of 7.6 K between the lowest and highest superheat temperature value (representing 31 % of the mean superheat temperature value), along with the standard deviation of 1.95 K . Both superheat temperature and critical heat flux deviations of represented data are well outside the measurement uncertainty of the experimental

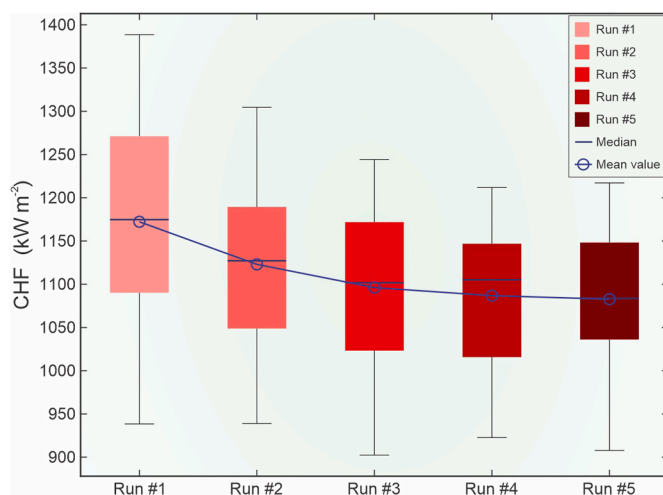


Fig. 6. The change of the critical heat flux by each subsequent experimental run.

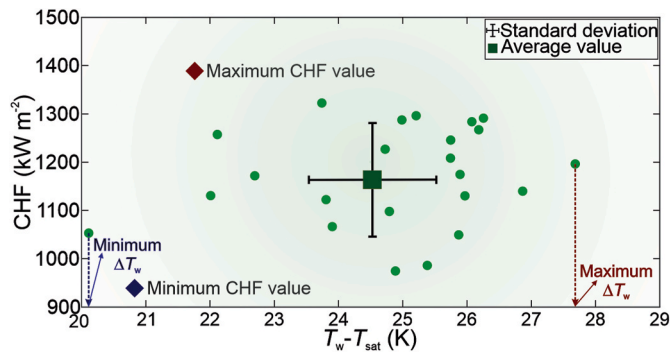


Fig. 8. CHF data scatter obtained during first experimental run.

apparatus. As aforementioned, we also can exclude the effect of roughness, material, and size of the sample on the final results. Although we exclude the important parameters that were shown in the literature that have an effect on CHF value, the scatter of the CHF and corresponding superheat temperature were still observed. This points towards an inherent scatter mechanism present in pool boiling CHF values of smooth copper surfaces, which cannot be avoided even with stringently controlled surface and experimental conditions.

4.4. Comparison with literature-reported data

In the course of research aimed at improving boiling heat transfer parameters, establishing a reference experiment as a baseline, to which the enhancements can be compared to, is crucial. Enhanced CHF values, achieved for example through surface engineering, are often reported as a relative improvement over the CHF value obtained on a reference (untreated) surface. However, the accuracy of this approach is limited by the large scatter of CHF data recorded on reference surfaces. This scatter has been observed in various studies reported in the literature, and it was also evident in our study. To address this limitation, we analyzed selected recent studies that reported relative CHF enhancement over the performance of their own reference surface. We calculated the relative enhancement over the minimum, average, and maximum CHF values as recorded on the reference surfaces in the present study to show by how much the reported relative enhancement can change due to the scatter of reference values. The results indicate that while other studies reported significant relative enhancement of CHF over their reference surfaces, there was no substantial enhancement or

there may even be a relative deterioration of CHF compared to the reference results obtained in the present study. Fig. 9 shows relative enhancement of maximum CHF recorded in each study over their own reference CHF and then over maximum, average and minimum CHF from the present study. CHF values used in Fig. 9 are shown in Table S4, Supporting Information, Section 7. Thus, reporting absolute enhancement values rather than relative values would be more accurate.

Additionally, our study establishes a comprehensive baseline for the complete boiling curve and critical heat flux. This baseline serves as a reference point for future comparisons and evaluations of enhancements over reference surfaces. This baseline is now also provided in the form of raw data that can be actively and accurately used by researchers in the future. The raw data of heat flux, and superheat temperature, and heat transfer coefficient of all 25 samples (5 runs per sample) for forming boiling curves are reported in Table S5 to Table S29 in Supporting Information, Section 8. In addition, the raw data of CHF with recorded superheat temperature at CHF are reported in Table S30, Supporting Information, Section 8. Incorporating this baseline enhances the safety and practicality of utilizing surfaces with improved boiling heat transfer characteristics in industrial applications. Thus, it is crucial to highlight the importance of reporting absolute improvements while emphasizing the inclusion of the measured reference CHF.

5. Conclusions

This study presents a thorough accessibility and analysis of 125 sets of data obtained on bare copper surfaces, exposed to saturated boiling of double-distilled water at amphoteric pressure. The results indicate that the scatter of CHF data is significant, with a standard deviation double that of the measurement uncertainty, suggesting that the latter alone cannot explain the observed scatter. The obtained maximum and minimum CHF are 1389 kW m^{-2} and 902 kW m^{-2} within 125 measured CHF values, repetitively, with the range of the scatter between them measuring 486 kW m^{-2} . The average recorded value of CHF is 1112 kW m^{-2} with a standard deviation of 102 kW m^{-2} . Furthermore, the maximum and minimum values of critical heat flux (CHF) recorded during the initial experimental run were determined to be 1389 kW m^{-2} and 939 kW m^{-2} , respectively, based on 25 CHF measurements. The observed maximal scatter between these values amounted to 450 kW m^{-2} . Thus, under identical samples, preparation procedure and boiling fluid properties, and measurement protocol, inherent scatter in CHF was evident not only during the first experimental run but also across all 125 measurements conducted. Experimental data indicate fit indicated good agreement with the Rohsenow's correlation, with a C_{sf} value of 0.0151. The study also showed that the number of times the surface was used for boiling tests had not a statistically significant impact on CHF ($p > 0.05$), while the number of experimental runs on each surface had a statistically significant impact on CHF ($p < 0.05$). Additionally, the data obtained in this study serves as an excellent baseline for enhancement evaluation in the future investigations of boiling heat transfer by providing an independent reference benchmark.

Associated content

Supporting Information

A detailed description of experimental methodology and measurement uncertainty; SEM and EDS analysis of tested surfaces; static contact angle and roughness measurements on surfaces before and after exposure to boiling tests; Evaluation of pool boiling heat transfer performance; Evaluation of the experimental data using Rohsenow's correlation; results of ANOVA test, fitting of experimental data to a normal distribution; additional information on data used from literature for comparison graph; detailed raw data of heat flux, temperature superheat and heat transfer coefficient recorded for 125 measurements recorded in this study.

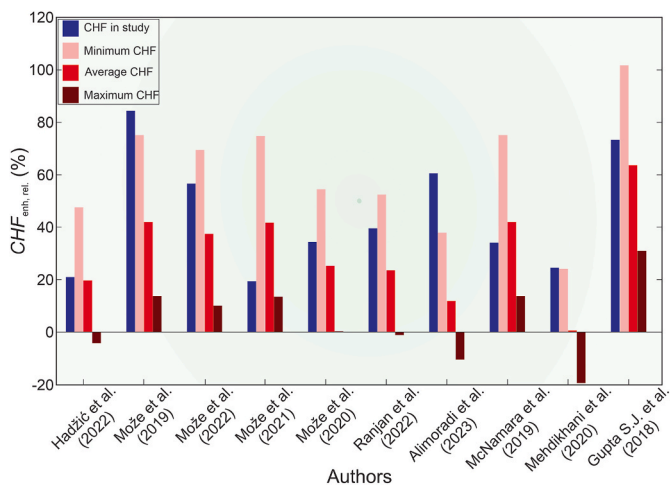


Fig. 9. Relative enhancement of maximum CHF from study over reference CHF from that study and then over maximum, average and minimum CHF from this study.

Nomenclature

Latin symbols	
C_p	isobaric specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)
C_{sf}	surface-fluid parameter
h	heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)
h	latent heat of vaporization (J kg^{-1})
g	gravitational acceleration (m s^{-2})
k	thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$)
m	exponent in Rohsenow equation
n	exponent in Rohsenow equation
$\dot{q}$	heat flux (W m^{-2})
R	arithmetical average value of roughness
T	temperature ($^{\circ}\text{C}$)
Greek symbols	
Δ	difference
μ	dynamic viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)
σ	surface tension (N m^{-1})
ρ	density (kg m^{-3})
a	average
enh	enhancement
G	gas
L	liquid
LG	liquid-gas
rel	relative
sf	surface/liquid
sat	saturated
w	wall
Abbreviations	
ANOVA	Analysis of variance
CHF	critical heat flux
BSE	backscattered electron
CVD	energy dispersive spectroscopy
HTC	heat transfer coefficient
ONB	onset of nucleate boiling
SE	secondary electron
SEM	scanning electron microscopy

Code availability

Code is available from the corresponding author upon request.

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CRediT authorship contribution statement

Armin Hadžić: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Matic Može:** Writing – original draft, Visualization, Project administration, Methodology, Formal analysis, Conceptualization. **Matevž Zupancič:** Writing – review & editing, Validation, Software, Resources, Formal analysis. **Iztok Golobič:** Writing – review & editing, Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All collected data is available in tabular form within the Supporting Information file. Code is available from the corresponding author upon request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijthermalsci.2024.109240>.

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